

eFinMoney — Product & System Documentation

Product	eFinMoney
Website	https://www.efin.money
Document type	Product & system documentation
Audience	Product, engineering, QA, compliance, partners
Version	1.0 — July 2026
Platform	Web application (responsive)

1. Introduction

1.1 Purpose of this document

This document describes what **eFinMoney** is, how it is structured, which capabilities are available to customers and staff, and how money movement is organized across corridors and providers. It is intended as a shared reference for understanding the product so teams (including QA) can design processes, reviews, and test coverage from a clear description of behaviour—not as a test plan itself.

1.2 Product summary

eFinMoney is a **multi-currency digital wallet and cross-border money platform**. Customers create accounts, complete identity verification, hold balances in multiple fiat currencies, fund those wallets, send money to bank accounts or mobile-money wallets, exchange currencies, manage payees, and (where enabled) use cards, bill payment, and related services.

The product is oriented toward **Africa ↔ Western** corridors (for example Nigeria, Ghana, Zambia, and CAD/USD/EUR/GBP), with a customer experience that presents **payment methods** under eFinMoney branding rather than naming underlying payment service providers.

1.3 Design principles

- **Multi-rail**: More than one funding or payout path may be offered for the same currency; the customer chooses a method.
- **White-label customer UI**: Customers see method labels such as *Express card*, *Pay by invoice*, *Card or bank transfer*, *Mobile money*, or *Interac e-Transfer*—not provider brand names on top-up and send choosers.
- **Feature gating**: Capabilities can be enabled or disabled via configuration without removing backend support.
- **KYC-first access**: Full use of the app shell is tied to identity verification progress.

2. System architecture (overview)

2.1 Technology stack

Layer	Technology
Frontend	Vite, React, TypeScript, shadcn/ui
Auth & database	Supabase (Auth, Postgres, Realtime)
Backend APIs	Supabase Edge Functions (Deno)
Hosting	Production web at efin.money; backend on Supabase

2.2 Logical components

- 1. **Marketing & legal site** — public pages (landing, features, privacy, terms, compliance).
- 2. **Customer application** — authenticated wallet, send, top-up, exchange, contacts, etc.
- 3. **Onboarding & KYC** — Persona-led identity; optional Sumsub for enhanced due diligence (admin).
- 4. **Admin / compliance portal** — staff tools for users, KYC, AML, finance, operations.
- 5. **Payment rails** — Edge Functions integrating PSPs for collection (top-up) and payout (send).
- 6. **Ledger / wallets** — per-currency wallet balances and transfer/journal history.

2.3 High-level money flow



3. Users and access

3.1 Customer users

Stage	Description
Registration	Email/password (Supabase Auth); confirmation and password reset supported
PIN	App PIN with reset flow
Onboarding	Welcome → identity verification → approved / enhanced / rejected paths
App access	Dashboard and money features after KYC gating rules are satisfied

3.2 Staff users

Staff access the **admin portal** (`/admin/*`) with role-based permissions (for example admin, finance, compliance). Capabilities include KYC review, user management, AML workflows, revenue/pricing views, support/communication, and API diagnostics.

3.3 Identity verification

Provider	Role
Persona	Primary customer identity verification during onboarding
Sumsub	Enhanced due diligence tools used from the admin KYC review context

Privacy and compliance pages describe customer-facing verification expectations; Sumsub is primarily an operations capability.

4. Customer application — modules

4.1 Dashboard

Home view of the customer’s financial activity: wallet snapshot and entry points to send, top up, exchange, and related actions.

Route: `/dashboard`

4.2 Wallets

Customers hold **one wallet per currency**. They can create wallets for supported currencies (commonly including NGN, USD, CAD, KES, GBP, EUR, GHS, UGX, TZS, ZMW, and others in the catalog), view balances, and open statements.

Capability	Route
Wallet list / create	<code>/wallets</code>
Statement	<code>/wallets/:walletId/statement</code>

4.3 Top up (fund wallet)

Customers fund a selected wallet via `/wallet/topup` . Available methods depend on the **wallet currency** and **feature configuration**. Multiple methods may appear for the same currency (multi-rail).

Customer-facing method labels

Label shown to customer	Typical currencies	Underlying rail (internal)
Express card	NGN; USD/EUR/GBP; CAD (charged as USD, credited CAD)	Nomba
Card or bank transfer	USD/EUR/GBP; CAD (via USD charge)	Fincra hosted checkout
Pay by invoice	USD, EUR, GBP, CAD	Paytota
Interac e-Transfer	CAD	Fincra Interac (when enabled)
Mobile money / Ghana MoMo	GHS	Ghana Pay
Mobile Money	ZMW (MTN, Airtel, Zamtel)	Elicate
Mobile money	XAF, KES, XOF, UGX	Swychr

Important product behaviour — CAD: Card/checkout rails that cannot charge CAD directly may **charge in USD** and **credit the CAD wallet**, with the quote UI explaining the conversion. Interac is a separate CAD-native collection path when that feature is enabled.

Important product behaviour — Fincra USD: For USD as charge currency, card is the primary method; bank transfer is not offered for USD on that rail. EUR/GBP may support card and bank transfer.

After payment, customers return via deposit/payment callback routes; balances update when verification or webhooks confirm success.

4.4 Send money

Customers send from a funded wallet to a beneficiary (bank or mobile money) via `/send`.

Live send corridors under default product configuration:

Corridor	Typical payout	Notes
Nigeria (NG)	Bank account	Primary international/Africa bank send path
Ghana (GH)	Mobile money or bank (as offered in UI)	Ghana Pay
Zambia (ZM)	Mobile money (MTN / Airtel / Zamtel)	Elicate

Other African corridors (for example Kenya, Uganda, Tanzania, Rwanda) may be offered when the *other African corridors* feature is enabled. Optional alternate payout toggles (for example UGX via Paytota) may appear when configured.

Send flows typically include: select corridor/beneficiary → amount → quote (fees/FX where applicable) → confirm → tracking under Transfers.

Related routes:

Route	Purpose
/send	Primary send
/transfers , /transfers/:id	History and tracking
/transactions/:journalId	Ledger detail
/transfers/canada	Canada domestic send (when enabled)
/send/cpn	Crypto/CPN send (when enabled)
/send/african-card	African card send (when Stripe features enabled)

Bank linking via **Plaid** may be available as a funding option on send when configured.

4.5 Receive money

Via `/wallet/receive` , customers can:

- View **in-network** receive details (account number / eFin tag)
- View **virtual account** information for supported currencies where provisioned
- Create **Zambia mobile-money payment links** (fixed or flexible amount) when the Zambia/Elicate capability is enabled—these are distinct from escrow “payment links” described below
- See informational guidance for Canada Interac where applicable (collection may be separate from domestic send)

4.6 Exchange

Fiat currency exchange between wallets is available at `/exchange` . Crypto trading surfaces are configuration-gated.

4.7 Contacts (payees)

Customers maintain a payee directory at `/contacts` (also `/payees`).

Categories: Person, Supplier, Employee, Contractor, Payee/Vendor, Other.

Directory fields include:

- Full name / business name
- Nickname (optional)
- Email (optional)
- Phone (optional; required when default payout is Mobile)
- Mailing address (optional): street, city, state/province, postal code
- Country
- Default payout method: None, EFT (Canada), Interac, Mobile, Bank
- Method-specific details (EFT institution/transit/account, Interac email, bank account details, mobile network as applicable)
- Notes and tags

Contacts can be used to start a send with prefilled beneficiary details.

4.8 Cards

When cards are enabled (`/cards`), customers can manage card products. Virtual cards via Swychr may be available when that rail is enabled. Stripe-based card funding and Connect surfaces are separately gated and may be off in the default product configuration.

4.9 Bill payment and airtime

When bill pay is enabled:

- `/pay-bills` — bill categories (airtime, cable, utilities, and others as catalogued)
- `/pay-bills/canada` — Canada bill pay
- Swychr airtime purchase may appear in the bills experience when Swychr is enabled

4.10 Payment links (escrow)

Escrow-style payment links (`/payment-links` , claim via `/claim/:code` , short links `/s/:code`) are a separate product surface from Zambia MoMo receive links. Escrow payment links are feature-gated and may show as coming soon when disabled.

4.11 Profile, security, support

Route	Purpose
<code>/more</code>	Settings hub
<code>/profile</code>	Profile settings
<code>/kyc</code>	KYC status / actions
<code>/security</code>	Security settings
<code>/support</code>	Support

4.12 Marketing and legal (public)

Route	Purpose
<code>/</code>	Landing
<code>/features</code> , <code>/how-it-works</code> , <code>/about</code> , <code>/contact</code>	Marketing
<code>/privacy</code> , <code>/terms</code> , <code>/compliance</code>	Legal and compliance disclosures
<code>/auth</code> , <code>/auth/confirm</code> , <code>/auth/reset-password</code>	Authentication
<code>/reset-pin</code>	PIN reset
<code>/portal</code>	Customer portal entry

5. Payment rails reference

This section documents **internal** rails for engineering and operations. Customer UI should continue to use the labels in §4.3.

5.1 Collection (top-up)

Rail	Currencies / notes	Edge function family (examples)
Nomba	NGN; international card; CAD via USD	<code>nomba-collection</code> , callbacks, banks, resolve
Fincra checkout	USD/EUR/GBP; CAD via USD charge	<code>fincra-initialize-checkout</code> , <code>fincra-verify-payment</code> , <code>fincra-webhook</code>
Fincra Interac	CAD Interac collections	<code>fincra-cad-interac</code> (+ webhook handling)
Paytota	Invoice for USD/EUR/GBP/CAD	<code>paytota-collection</code> , <code>paytota-webhook</code>
Ghana Pay	GHS MoMo	<code>ghana-collection</code> , callbacks
Elicate	ZMW MoMo	<code>elicate-charge</code> , <code>elicate-checkout</code> , <code>elicate-webhook</code> , <code>status/stream</code>
Swychr	XAF/KES/XOF/UGX payin	<code>swychr-collection</code> , payin verify/webhook
Flutterwave	Legacy / gated	<code>flw-*</code>

5.2 Payout (send)

Rail	Typical use
Nomba	Nigeria bank payouts; FX/conversion related APIs
Ghana Pay	GHS MoMo/bank payouts
Elicate	ZMW MoMo payouts (<code>elicate-payout</code>)
Paytota	Optional UGX MoMo payout
Fincra	Optional African payout toggle where configured
Swychr	Payout / NGN fallback where configured
Paysafe / Interac	Canada domestic (when enabled)
Flutterwave	Legacy bills and historical payout paths

5.3 Ancillary rails

Capability	Notes
Plaid	Link external bank accounts
Swychr cards	Virtual card issue and operations
Swychr airtime	Catalog and recharge
Elicate payment links	ZMW MoMo receive links
Stripe / Adyen / Circle / Stellar	Present in backend; many customer surfaces gated off by default

5.4 Multi-rail policy

Adding a new rail must **not** remove an existing live rail for the same currency. The customer chooses among available methods. Exclusivity (forcing a single provider) is not the product rule.

6. Feature configuration

Product visibility is controlled in `src/lib/productFeatures.ts`, overridable with `VITE_FEATURE_*` environment variables.

Feature key	Default (July 2026)	Effect when enabled
nombaNigeria	On	NGN Express card; Nomba send/FX paths
ghanaPay	On	GHS MoMo top-up and send
elicate	On	ZMW MoMo top-up, send, receive links
paytota	On	Pay by invoice for Western currencies
paytotaPayout	On	UGX MoMo payout option on Send
fincra	On	Card or bank transfer top-up
fincraInterac	Off	CAD Interac e-Transfer top-up
swychr	On	Swychr top-up, cards, airtime surfaces
plaid	On	Bank link on send
billPay	On	Bill pay routes
cards	On	Cards route
canadaDomestic	Off	Canada domestic transfers
paymentLinks	Off	Escrow payment links
flutterwave	Off	Flutterwave top-up UI
otherAfricanCorridors	Off	Additional Africa send corridors live
crypto	Off	Crypto send/exchange surfaces
stripe	Off (hardcoded)	Stripe Connect / African card send
adyen	Off	Adyen surfaces

Disabled features typically show a **Coming soon** experience rather than a broken page.

7. Admin and compliance portal

Staff use `/admin/login` and the admin layout. Major areas:

Area	Examples of routes / purpose
Dashboard	/admin/dashboard , board dashboard
KYC	Queue and review; KYC config; risk tiers
Users & staff	User list/detail; staff onboarding and training
Finance	Finance, revenue, pricing
Operations	Operations hub, incidents, settlement reconciliation
Communication	Communication hub, support inbox
AML / compliance	AML policy, CDD, EDD, sanctions, PEP, STR, transaction monitoring, trade AML
Regulatory	LCTR, EFTR, wire transfers, travel rule, geographic risk, correspondent banking
Controls	Period-end controls, evidence repository, regulatory changes
Security	Security monitoring, operational risks
Diagnostics	System diagnostics, audit log, API management, data export
Payments (ops)	Adyen admin links/transactions where used

Admin screens may display provider names for operational clarity; the white-label rule applies to **customer** money-movement UI.

8. End-to-end product journeys (narrative)

8.1 New customer

1. Discovers eFinMoney via marketing site.
2. Registers and confirms authentication.
3. Completes onboarding and Persona identity verification.
4. On approval, accesses dashboard and creates one or more currency wallets.

8.2 Fund and send (Africa)

1. Opens Top up, selects e.g. NGN, GHS, or ZMW wallet.
2. Completes the presented mobile money or card method.
3. Balance updates after confirmation.
4. Adds or selects a contact (bank or MoMo).
5. Sends via Send; tracks status under Transfers.

8.3 Fund CAD / USD wallet

1. Selects CAD or USD (or EUR/GBP) wallet.
2. Chooses among Express card, Card or bank transfer, and/or Pay by invoice (as configured).

3. For CAD card paths, may pay in USD equivalent and receive CAD credit.
4. Optionally uses Interac when that feature is turned on.

8.4 Receive in Zambia

1. Opens Receive.
2. Creates a MoMo payment link (fixed or open amount).
3. Payer completes mobile money; wallet is credited when the provider confirms.

8.5 Exchange

1. Holds two fiat wallets.
2. Uses Exchange to convert at the quoted rate/fees.
3. Both balances update.

9. Operational notes and known constraints

These are product/environment realities that affect how the system behaves in production:

1. **CAD card top-up** often uses a USD charge currency with CAD wallet credit.
2. **Fincra Interac** is implemented but may be off by default until alias/credentials and flag are ready.
3. **Merchant enablement** at the PSP (e.g. USD card processing) can block otherwise valid customer cards.
4. **Canada domestic send** and **escrow payment links** may be coming soon while Zambia MoMo receive links remain available.
5. **Nomba upstream credentials** (e.g. JWT) can break payouts while directory/lookup APIs still succeed.
6. **Local development redirects** after hosted checkout may return to the configured production app URL.
7. Prefer `productFeatures.ts` over older markdown rail lists when determining what is live in the UI.

10. Glossary

Term	Definition
Wallet	Customer balance in a single currency
Corridor	Country/currency path for sending money
Rail / PSP	Payment service provider powering collection or payout
MoMo	Mobile money
White-label	Customer sees eFinMoney method names, not PSP brands
Edge Function	Serverless backend API on Supabase
KYC / CDD / EDD	Know Your Customer / Customer Due Diligence / Enhanced Due Diligence
Multi-rail	Multiple providers offered for the same currency or flow

11. Document control

Item	Detail
Product name	eFinMoney
Primary code references	<code>src/lib/productFeatures.ts</code> , <code>src/lib/walletTopupGateway.ts</code> , <code>src/App.tsx</code> , <code>top-up/send/receive/contacts</code> pages, <code>supabase/functions/*</code>
Related internal notes	<code>docs/PRODUCT_RAILS.md</code> , <code>docs/swychr/README.md</code>
Update when	Feature-flag defaults change, new corridor goes live, or customer-facing method labels change